

SW Team - CMake Guidelines

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Purpose

This page is a collection of **informal** guidelines for handling CMake files.

Despite being informal, there should be good reasons for not following the guidelines.

Rules

cmake.version1

- Every CMakeLists.txt should specify the minimum required version with e.g. `cmake_minimum_required(VERSION 1.2)`.
- The major.minor version (e.g. 1.2) should be used to specify a CMake version unless a comment explains why the major.minor.patch version (e.g. 1.2.3) is required.
- *Rational: CMake feature set is defined by major.minor version.*

cmake.version2

- **Created new CMakeLists.txt**
 - A new created CMakeLists.txt should simply require the **current CMake version of the build container**.
 - *Rational: We do not want to analyse which CMake version satisfies CMake features used in the new file.*
- **Changed existing CMakeLists.txt**
 - If a new feature is used, the version should be updated to the **current CMake version of the build container**.
 - *Rational: We do not want to analyse which CMake version satisfies CMake features used in the new file.*
- **Copied existing CMakeLists.txt**
 - A copied CMakeLists.txt should keep the **required CMake version of the original file**.
 - *Rational: Keep consistency within a group of similar files.*

cmake.file-structure

- CMake project files CMakeLists.txt should be added to another project with `add_subdirectory()`.
- File with CMake definitions (functions, variables) should have the suffix `.cmake` and be included with `include()`.